BIONEX

Hybrid BCI-Controlled Robotic Prosthetic Arm

Engineering Final Project: BCI and ESP32 Based

PROBLEM STATEMENT & OBJECTIVE

THE PROBLEM

Advanced prosthetics are expensive (>$10,000) and often inaccessible.

Existing low-cost myoelectric (EMG-only) solutions lack the **adaptability** needed for complex tasks, requiring awkward muscle commands for mode-switching.

THE OBJECTIVE

To develop a low-cost, 3D-printed bionic arm (Inmoov-based) using a **hybrid brain-computer interface (BCI)**.
The system fuses:

- **EMG:** Proportional control for grasping force.
- **EEG:** Cognitive commands (attention, eye blinks) for mode/gesture switching.

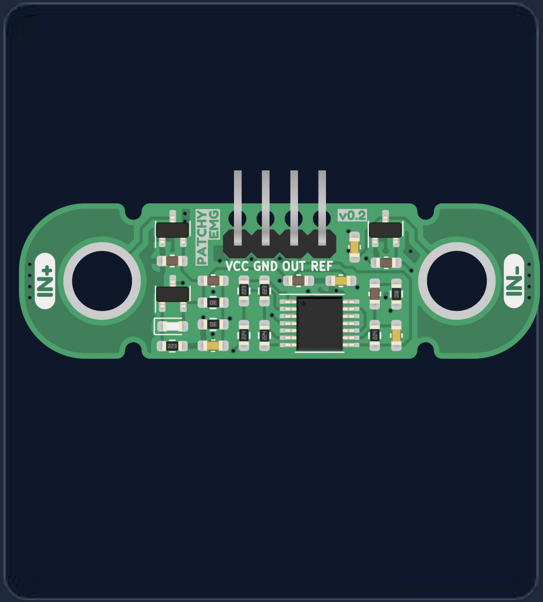
CORE HARDWARE COMPONENTS

ESP32 MICROCONTROLLER

Processing for real-time sensor data and control.

EEG SENSOR (BCI)

Reads cognitive signals for gesture mode switching.



BIOAMP PATCHY (EMG)

Detects muscle potentials for grasping action.

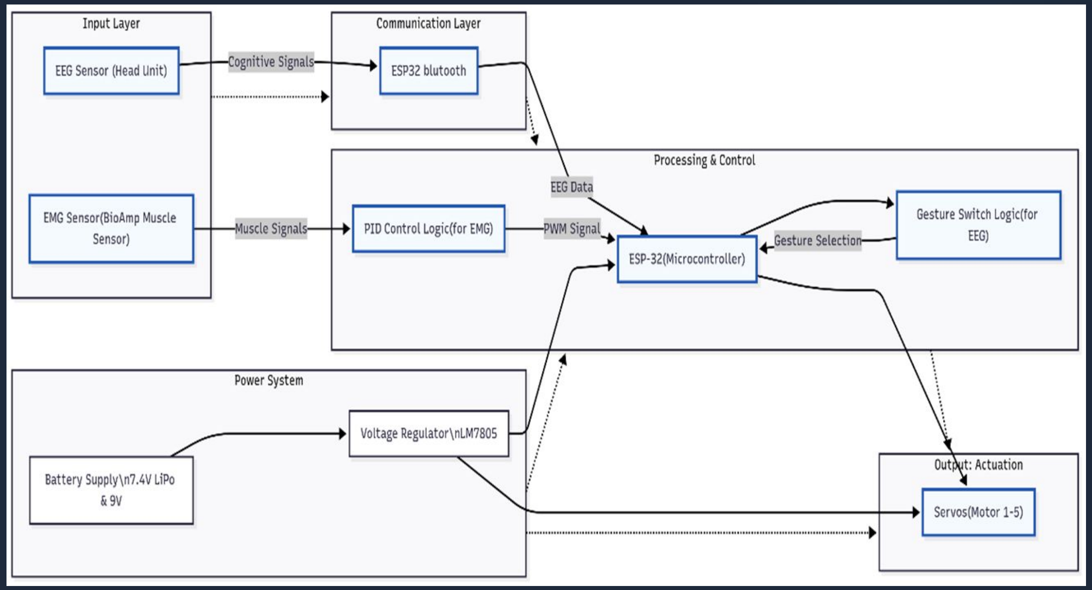
MG90s SERVOS (X5)

High-torque actuators for finger control and force.

SYSTEM ARCHITECTURE: BIONEX HYBRID CONTROL

ARCHITECTURE LAYERS

- **Input Layer:** EMG (Muscle) and EEG (Cognitive) signals are acquired.
- **Communication Layer:** Bluetooth links EEG data to the ESP32.
- **Processing & Control:** ESP32 handles dual logic and actuation.
- **Output Layer:** 5 Servos control the 3D-printed Inmoov hand structure.



EMG & ADC SIGNAL CONDITIONING

EMG FOR PROPORTIONAL CONTROL

EMG amplitude is directly proportional to the force of muscle contraction.

$$\text{Grasping Force} \propto \text{EMG Amplitude}$$

The system translates varying muscle intensity into a smooth range of servo positions, allowing for nuanced grasping.

ADC SAFETY & NOISE REDUCTION

The analog EMG output is conditioned for the ESP32's 3.3V ADC input using a **voltage divider (10kΩ/22kΩ)**.

Digital filtering (moving average) is crucial to reduce high-frequency muscle noise and ensure a stable control signal.

DUAL SIGNAL PROCESSING PIPELINE

EMG PROCESSING (ACTION)

Goal: Proportional control of grasp.

- Rectification & Envelope Generation
- PID Control Logic (for smooth force application)
- Output: Continuous servo position angle.

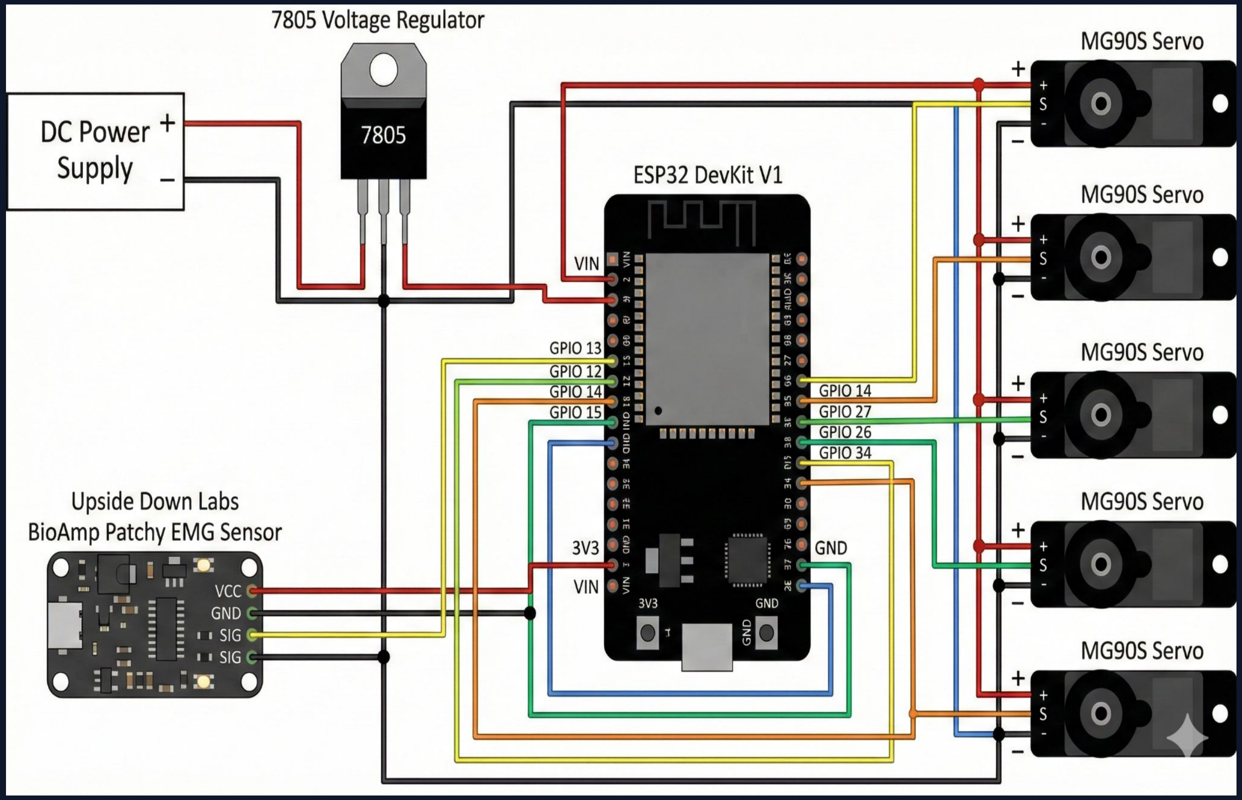
EEG PROCESSING (SWITCHING)

Goal: Switch between programmed gestures.

- Attention/Meditation Levels from BCI data.
- Blink detection for instantaneous switching.
- Output: Discrete State Change (e.g., from Pinch Mode to Power Grasp Mode).

CIRCUIT & MECHANICAL INTEGRATION

- **Mechanical Base:** The hand and forearm were constructed using **3D-printed parts** based on the open-source Inmoov project.
- **Power Safety:** A powerful **external 5V PSU (3-5A)** is mandatory for the 5 MG90s servos.
- **Common Ground:** Essential link between the external PSU and the ESP32 for stable control.
- **Actuation:** 5 servos mapped to 5 distinct ESP32 LEDC PWM channels.



ACTUATION: PWM AND SMOOTH MOTION

LEDC HARDWARE PWM

The ESP32's LEDC peripheral generates stable, high-resolution PWM signals (10-12 bit) necessary for precise and smooth servo positioning.

ANGLE STEPPING

Instead of instantaneous movement, the code uses **incremental angle steps**.

- The servo target angle is reached gradually.
- Result: Fluid, natural motion, preventing abrupt mechanical wear.

SERIAL MOTOR CONTROL MODE

A secondary mode implemented via serial commands (e.g., s1 90, all) 180) allows for:

- Precise servo calibration and trimming.
- Debugging complex movement sequences.
- Manual testing of the high-torque MG90s servo motors.

SOFTWARE IMPLEMENTATION & ALGORITHM

CORE ALGORITHM

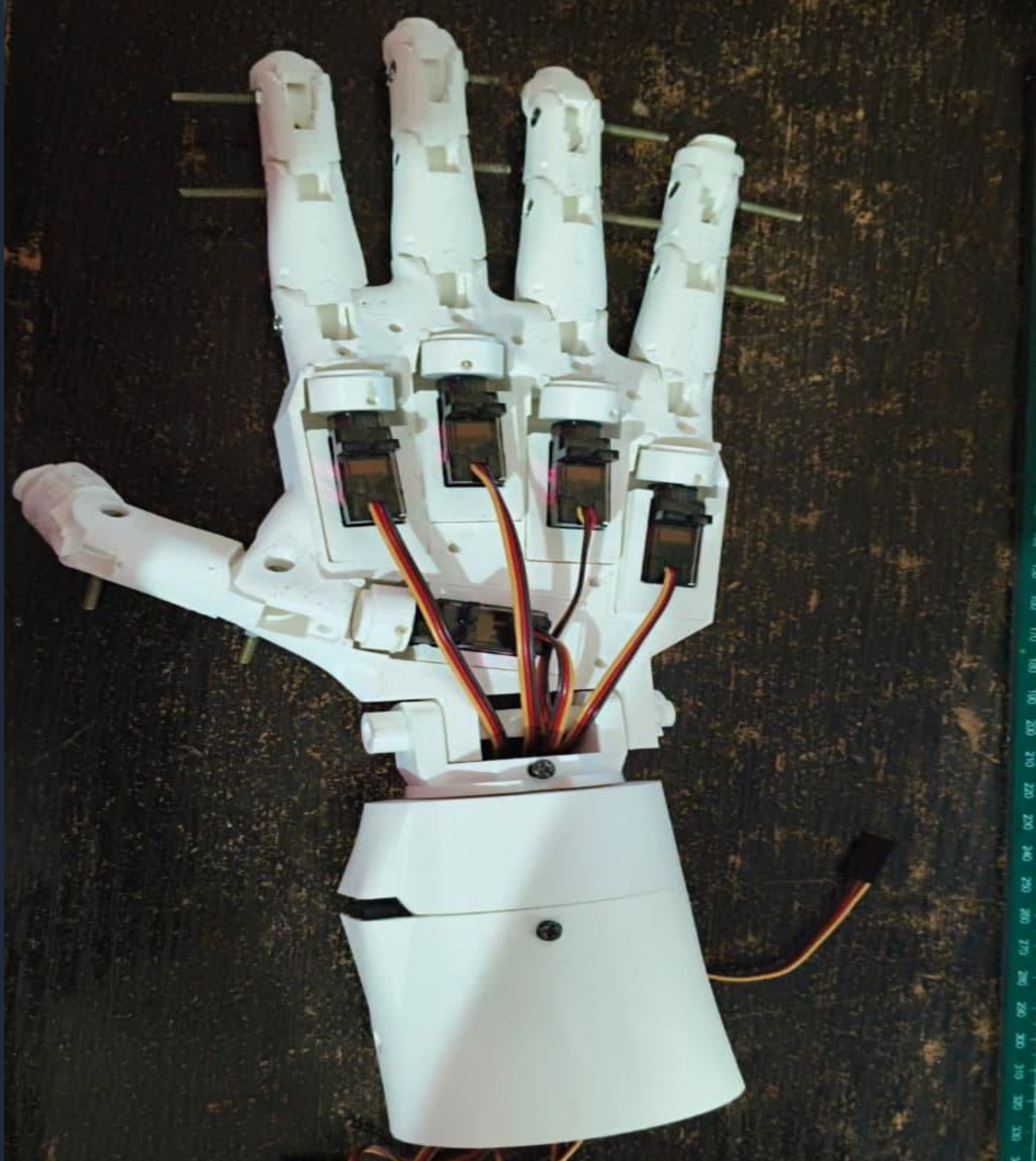
The system is governed by a state machine:

- **Mode Determination:** EEG data sets the current hand gesture/state (e.g., Pinch, Relax, Power).
- **Execution:** EMG data determines the proportional force/position within that selected state.
- **Calibration:** Dynamic thresholds (Relax/Flex) are established upon startup.

```
// Simplified Servo Update Logic for Proportional Control void updateServoPosition (int emgVal, int targetPos) {
// Map filtered EMG (e.g., 500-4095) to Servo Angle (0-180) int angle = map(emgVal, 500, 4095, 0, 180);
// Limit movement to prevent mechanical lock angle = constrain(angle, 0, 180);
// Smooth transition using angle steps
incrementallySetServo (angle); } void loop() { int emgValue =
readAndFilterEMG (); int currentMode =
getGestureMode (); // Based on EEG/BCI
// 2. Execute Action based on Mode and EMG
executeModeAction (currentMode, emgValue);
delay(20); }
```

RESULTS & VALIDATION

- **Hybrid Control:** Successful demonstration of **EEG mode switching** (e.g., blink to change grip) and **EMG proportional grasping**.
- **Reliability:** Anti-chatter logic and dynamic calibration ensure stable operation.
- **Mechanical:** The 3D-printed Inmoov structure provided the low-cost, modular base required.
- **Performance:** High-torque MG996R servos provided sufficient force for grasping common objects.



FUTURE SCOPE

SENSORY FEEDBACK

Integrate pressure/haptic sensors in fingertips to provide real-time force feedback to the user.

MINIATURIZATION

Develop a custom, shielded PCB to integrate all components (EMG/BCI receiver) into a small, wearable form factor.

LIPO POWER & MANAGEMENT

Develop a robust, integrated LiPo battery management system (7.4V) for full portability and extended use.

CONCLUSION

BIONEX successfully validates the concept of a hybrid BCI prosthetic arm that is both highly functional and economically viable.

The fusion of EMG for proportional action and EEG for cognitive mode-switching provides a more intuitive and versatile control scheme than traditional myoelectric devices.

THANK YOU

Questions?

github.com/IAbhisek/BIONEX